Studying Augmentation-Noise Alignment in Self-Supervised Learning on Medical Imaging

Coulibaly Zié[1], Daniel Thomas[1], Mohamed El Bechir Mohamed[1], Verier Antoine[1], and Xhepi Nardi[1]

[1]Institut Polytechnique de Paris

March 29, 2026

Abstract

Self-supervised learning (SSL) has become a dominant paradigm for learning visual representations without labels. Recent theoretical work by Van Assel et al. [1] reveals that the *alignment* between data augmentations and irrelevant noise governs SSL performance: joint-embedding methods require weaker alignment than reconstruction-based methods, making them inherently more robust. Although their empirical validation focuses on natural images, medical imaging modalities exhibit fundamentally different noise distributions and acquisition artifacts. To investigate whether these alignment principles generalize to clinical domains, we extend this framework to medical imaging using OrganAMNIST from MedMNIST-C [11], comparing both reconstruction and joint-embedding principles under controlled corruption. We implement a continuous pixel-level residual technique that controls alignment strength α, sweeping it over a fine grid to empirically identify each method's critical alignment threshold and determine which SSL paradigm is most robust to clinical distribution shifts.

1 Introduction

A wide variety of self-supervised learning (SSL) frameworks have been developed to extract meaningful visual representations without relying on manual labels. While these approaches have demonstrated remarkable success on standard natural image benchmarks [3, 6, 9], they fundamentally diverge in how they build these representations.

Specifically, *reconstruction-based methods* (e.g., Masked Autoencoders [2]) learn by predicting missing or corrupted pixel-level information from an input. In contrast, *joint-embedding architectures* (e.g., SimCLR [3], VICReg [4]) operate by maximizing the similarity between differently augmented views of the same image within an abstract latent space. As these models are increasingly deployed in high-stakes clinical settings, a critical question arises: *why are certain SSL architectures inherently more robust to noise and clinical artifacts than others?*

Recent theoretical work by Van Assel et al. [1] addresses this through the lens of *augmentation-noise alignment*—a measure of how well an augmentation strategy correlates with the underlying irrelevant noise distribution in the data.

However, this theoretical framework and its empirical validations are currently restricted to natural image datasets (e.g., CIFAR-10, ImageNet-C) using additive noise models. Given the complex structures with non-additive noise profiles inherent to clinical image acquisition, it remains unclear if these robustness guarantees translate to healthcare applications. First, we look into **domain generalization** by evaluating the OrganAMNIST dataset [10] from MedMNIST-C [11] to determine if the critical alignment threshold α^* and the theoretical advantage of joint-embedding methods persist on complex medical anatomies. Second, to address the gap between theoretical **continuous optimization** and discrete evaluations, we explore a pixel-level corruption blending technique. By sweeping this real-valued parameter over a fine-grained grid, we empirically approximate the critical threshold α. Third, we challenge the theory's **additive noise assumption** mentions in [1] with non-additive Gaussian blur, effectively stress-testing the

framework against realistic, convolutive corruptions. Finally, we investigate the **information-theoretic ceilings** imposed by such corruptions; as blur irreversibly destroys high-frequency spatial information, we examine whether optimizing alignment remains beneficial and if joint-embedding architectures can maintain their performances when pushed to extreme data loss.

Contributions. In summary, our core contributions are as follows:

- **Domain extension to Medical Imaging:** We translate the augmentation-noise alignment framework from natural images to the medical domain, systematically comparing reconstruction (MAE) and joint-embedding (SimCLR, VICReg, LeJEPA) methods on OrganAMNIST under realistic clinical corruptions.
- **Evaluating non-additive corruptions:** We stress-test the theoretical assumptions of strictly additive noise by evaluating both additive (Gaussian noise) and non-additive (Gaussian blur) corruptions to observe information-theoretic ceilings.
- **Empirical limits of Joint-Embedding:** We provide novel empirical evidence that while the joint-embedding paradigm allows for superior noise filtering (e.g., SimCLR), methods lacking negative sampling heuristics (such as VICReg and LeJEPA) are highly susceptible to representation collapse under anatomical corruption.

2 Background

2.1 Self-Supervised Learning

SSL trains an encoder $f_\theta : \mathbb{R}^n \to \mathbb{R}^d$ with $d \ll n$ on unlabeled data via a pretext objective; learned representations are then usually evaluated by *linear probing*. Two paradigms exist, with extended descriptions in Section A:

Reconstruction-based SSL

The encoder maps a corrupted or masked input to a latent space, and a decoder attempts to recover the original signal. To minimize reconstruction error, the encoder is forced to retain a high-fidelity representation of the input, including low-level details and noise that may not be relevant to downstream semantic tasks.

Joint-Embedding SSL

Two differently augmented versions of the same image are processed by an encoder. The objective is to maximize the similarity between their embeddings in a high-dimensional latent space. Because there is no pixel-level reconstruction, the encoder learns to be invariant to the transformations applied, discarding ”nuisance variables” (like lighting or cropping) in favor of core semantic features.

In this work, we compare the following self-supervised learning architectures:

- **MAE** [2]: reconstruction-based learning that masks a high proportion of input patches and uses an asymmetric encoder-decoder architecture to reconstruct missing pixels in the original pixel space.
- **SimCLR** [3]: a joint-embedding method, using a contrastive learning with the NT-Xent loss, pulling positive pairs together and pushing negatives apart.
- **VICReg** [4]: a joint-embedding architecture that avoids negatives by regularizing embedding variance and decorrelating dimensions via invariance, variance, and covariance terms.
- **LeJEPA** [7]: a joint-embedding predictive architecture. Instead of relying on architectural tricks (like stop-gradients) to prevent the model from outputting the same vector for every image (representation collapse), it uses Sketched Isotropic Gaussian Regularization (SIGReg) to mathematically ensure the embeddings remain diverse and well-distributed.

2.2 Augmentation-Noise Alignment

Data augmentation is the primary mechanism through which SSL learns what to discard: if a transformation varies between views but both depict the same image, the destroyed information must be semantically irrelevant. Standard augmentations like crop, flip, and affine transforms were designed for natural photographs and do not account for domain-specific noise such as scanner artifacts.

Consider data $x = s + \varepsilon$ where s is signal and ε is irrelevant noise. If ε is identical across both SSL views, the encoder has no incentive to discard it.

The Alignment Principle

If we inject additional noise of the *same type* independently per view, the noise component now *varies* between views while the semantic content remains shared. The SSL objective then forces the encoder to ignore the noise altogether.

Formally, given a view $v = \text{StandardAug}(x)$ and corruption residual $\gamma = \text{corrupt}(v) - v$, the aligned view is:

$$\tau(x) = v + \alpha \cdot \gamma, \quad \alpha \geq 0. \tag{1}$$

At $\alpha = 0$ the augmentation is *misaligned*; increasing α strengthens alignment with the noise.

Van Assel et al. formalize this in a linear model $x = Us + V\varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2 I)$:

Theorem 1 (Alignment Threshold [1, Theorem 3.5])**.** *For both reconstruction (RC) and joint-embedding (JE) methods, there exist critical thresholds α^*_{RC} and α^*_{JE} such that: (i) $\alpha > \alpha^*$ achieves optimal Bayes downstream accuracy; (ii) $\alpha < \alpha^*$ allows noise leakage. Furthermore, $\alpha^*_{JE} \leq \alpha^*_{RC}$, with strict inequality when σ_ε is large.*

The ordering reflects that reconstruction methods must preserve pixel-level detail including noise, while joint-embedding methods only need latent agreement and can discard noise more easily.

Scope of the theoretical guarantees

The formal results of [1], specifically Propositions 4.3, 4.4 and Corollary 4.5, are proved under the assumption that both the data corruption and the augmentation are **additive and independent** Gaussian perturbations: $\tau(x) = x + \theta + \alpha\gamma$ with $\theta \sim \mathcal{N}(0, \Theta)$ and $\gamma \sim \mathcal{N}(0, \Gamma)$. The paper itself notes that this excludes corruptions where data and noise are intertwined, such as spatter, and indeed reports that noise injection is less effective for such cases. Non-additive corruptions, such as Gaussian blur (a convolution) or pixelation (a lossy downsampling), fall outside the formal guarantees. Part of our experimental contribution is precisely to test whether the alignment principle still holds empirically beyond the additive regime.

3 Experimental Setup

3.1 Dataset and Corruptions

We evaluate on OrganAMNIST [10] (64×64 grayscale CT images; 34k train / 17k test). Following MedMNIST-C [11], we inject four corruption types at five severities using a fast GPU-accelerated pipeline. To prevent scaling artifacts, corruptions are applied at the native 224x224 resolution before down sampling to 64x64. Corrupted images are generated offline and stored statically, ensuring each sample has a fixed noise component ϵ across all training epochs to align with our theoretical framework.

3.2 Corruption-Aligned Augmentation

Following [1], each SSL view undergoes standard augmentations: random resized crop with scale 0.8–1.0, horizontal flip, and random affine with rotation $\leq 15°$ and translation $\leq 10\%$. When $\alpha > 0$, we compute

the corruption residual γ and blend it into the view via Equation (1). For additive corruptions like Gaussian noise, γ is a random sample from $\mathcal{N}(0, \sigma^2)$; for non-additive ones like blur and pixelation, γ captures the pixel difference between corrupted and clean views. We sweep $\alpha \in \{0.0, 0.1, 0.5, 1.0, 3.0, 5.0\}$.

3.3 Training and Evaluation

We pre-train a small Vision Transformer (ViT) using four self-supervised methods (SimCLR, VICReg, MAE, and LeJEPA), followed by a frozen linear probe to evaluate top-1 classification accuracy on clean and corrupted OrganAMNIST datasets. Full architectural configurations and training hyperparameters are detailed in Appendix B.

4 Results

	SimCLR		**MAE**		**VicReg**		**LeJepa**	
α	**Clean**	**Corrupted**	**Clean**	**Corrupted**	**Clean**	**Corrupted**	**Clean**	**Corrupted**
0.0	79.0	82.2	70.1	73.7	49.8	50.0	**41.3**	**40.4**
0.1	79.7	82.2	71.0	72.9	50.2	50.0	40.1	39.3
0.5	80.5	**82.4**	**75.3**	**76.8**	48.5	48.0	36.0	35.1
1.0	**80.9**	**82.4**	74.4	76.2	49.5	49.3	34.7	34.5
3.0	78.9	81.6	72.4	74.4	**50.5**	**50.8**	36.2	35.7
5.0	78.1	80.0	72.8	74.0	46.9	47.7	35.7	34.9

Table 1: OrganAMNIST classification accuracy with clean test set and Gaussian noise corruption (severity level 3) as a function of alignment strength (α).

	SimCLR		**MAE**	
α	**Clean**	**Corrupted**	**Clean**	**Corrupted**
0.0	80.7	81.5	74.9	78.1
0.1	80.1	81.6	75.1	78.0
0.5	**82.0**	**83.1**	75.6	79.9
1.0	80.1	82.0	**78.3**	**81.8**
3.0	78.8	82.1	72.8	76.2
5.0	80.4	81.9	73.2	76.5

Table 2: OrganAMNIST classification accuracy with clean and Gaussian blur corrupted (severity level 3) test sets as a function of alignment strength (α).

4.1 Comparative Analysis of Alignment Sensitivity

The results presented in Table 1 and Table 2 provide empirical evidence that Joint-Embedding (JE) and Reconstruction-based (RC) methods exhibit distinct sensitivities to augmentation-noise alignment. Consistent with theoretical predictions, SimCLR (a JE method) demonstrates superior robustness across the alignment spectrum. In the Gaussian noise regime, SimCLR maintains approximately 82.4% accuracy with minimal variation as α increases, whereas MAE (an RC method) exhibits a distinct performance curve, requiring moderate alignment ($\alpha = 0.5$) to reach its peak (76.8%). This confirms that JE methods impose a strictly weaker alignment condition to achieve optimality. The performance of MAE is notably more volatile.

At $\alpha = 0$, MAE suffers from "noise leakage" because the reconstruction objective forces the model to capture low-level, variance-explaining features, including irrelevant noise artifacts. While increasing

α provides the necessary signal for the encoder to disregard these components, excessive alignment ($\alpha = 5.0$) results in a performance drop to 74.0% in Table 1. This suggests that over-alignment may cause the model to discard shared semantic anatomical features in favor of noise invariance.

The observed failure of VICReg and LeJEPA (achieving ∼50% and ∼41% respectively) warrants specific discussion. While the JE paradigm mathematically allows for superior noise filtering, our results suggest that these specific implementations may be highly sensitive to hyperparameter configurations or training schedules when applied to anatomical structures. The superior performance of SimCLR suggests that explicit negative sampling may provide a more robust training signal for medical imaging when compared to the regularization-based collapse prevention in VICReg and LeJEPA, especially under the constraint of a fixed, relatively shallow backbone.

4.2 Robustness and Severity Scaling

	SimCLR						**MAE**					
	Clean			**Corrupted**			**Clean**			**Corrupted**		
α	**Sev 1**	**Sev 3**	**Sev 5**	**Sev 1**	**Sev 3**	**Sev 5**	**Sev 1**	**Sev 3**	**Sev 5**	**Sev 1**	**Sev 3**	**Sev 5**
0.0	**82.4**	**80.7**	**78.9**	82.7	81.5	**82.0**	77.7	74.9	71.3	79.2	78.1	**78.6**
1.0	81.8	80.1	77.3	**82.8**	**82.0**	81.4	**78.4**	**78.3**	69.1	**81.1**	**81.8**	76.4
5.0	81.1	80.4	76.7	81.7	81.9	81.1	77.6	73.2	**74.0**	79.8	76.5	77.8

Table 3: OrganAMNIST classification accuracy with Gaussian blur corruption at different severity levels (1, 3, 5) as a function of alignment strength (α).

Table 3 examines the scalability of the alignment principle across varying corruption severities. SimCLR exhibits stability, maintaining a corrupted accuracy of 82.0% at Severity 5 even with $\alpha = 0.0$. The addition of explicit alignment ($\alpha = 1.0$) does not yield significant gains, confirming that JE methods inherently prioritize global context over raw perceptual fidelity.In contrast, MAE benefits from alignment at lower severities (Sev 1 and 3) but faces a "ceiling effect" at Severity 5, where performance with $\alpha = 1.0$ (76.4%) falls below the baseline (78.6%). We hypothesize that at extreme blur severities, the noise becomes non-separable from the anatomical signal in the pixel space.

5 Discussion

Empirical results confirm the alignment framework: SimCLR (JE) demonstrates superior robustness over MAE (RC) by requiring a strictly lower alignment threshold α. Unlike MAE, which captures high-variance noise for pixel-level recovery, JE paradigms effectively discard nuisance variables in latent space. This advantage persists for non-additive corruptions, where RC methods suffer from signal-noise inseparability. Although VICReg and LeJEPA underperform due to hyperparameter sensitivity on fixed backbones, the overall stability of SimCLR suggests that contrastive methods offer more reliable training signals in clinical domains than current regularization-based alternatives.

6 Conclusion

This study confirms that augmentation-noise alignment generalizes to medical datasets, with joint-embedding architectures providing a significantly wider safety margin against clinical noise. Due to a lower critical threshold α^*, JE methods reduce the necessity for precise augmentation tuning across varying scanner artifacts. Given the fixed architectural capacity used in this study, future work should focus on automated, data-driven alignment optimization to navigate the unpredictable noise profiles inherent in real-world clinical deployment

References

[1] H. Van Assel, M. Ibrahim, T. Biancalani, A. Regev, and R. Balestriero. Joint-embedding vs reconstruction: Provable benefits of latent space prediction for self-supervised learning. *arXiv preprint arXiv:2505.12477*, 2025.

[2] He, Kaiming, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. "Masked autoencoders are scalable vision learners." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 16000–16009. 2022.

[3] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton. A simple framework for contrastive learning of visual representations. In *Proc. ICML*, 2020.

[4] A. Bardes, J. Ponce, and Y. LeCun. VICReg: Variance-invariance-covariance regularization for self-supervised learning. In *Proc. ICLR*, 2022.

[5] Balestriero, Randall, and Yann LeCun. "LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics." *arXiv preprint arXiv:2511.08544*, 2025.

[6] M. Caron, H. Touvron, I. Misra, H. Jégou, J. Mairal, P. Bojanowski, and A. Joulin. Emerging properties in self-supervised vision transformers. In *Proc. ICCV*, 2021.

[7] R. Balestriero and Y. LeCun. LeJEPA: Provable and scalable self-supervised learning without the heuristics. *arXiv preprint arXiv:2511.08544*, 2025.

[8] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick. Masked autoencoders are scalable vision learners. In *Proc. CVPR*, 2022.

[9] J.-B. Grill, F. Strub, F. Altché, C. Tallec, P. Richemond, E. Buchatskaya, C. Doersch, B. Avila Pinto, Z. Zheng, M. Azabou, et al. Bootstrap your own latent: A new approach to self-supervised learning. In *Proc. NeurIPS*, 2020.

[10] J. Yang, R. Shi, D. Wei, Z. Liu, L. Zhao, B. Ke, H. Pfister, and B. Ni. MedMNIST v2: A large-scale lightweight benchmark for 2D and 3D biomedical image classification. *Scientific Data*, 10(1):41, 2023.

[11] F. Di Salvo, S. Doerrich, and C. Ledig. MedMNIST-C: Comprehensive benchmark and improved classifier robustness by simulating realistic image corruptions. *arXiv preprint arXiv:2406.17536*, 2024.

A SSL Methods: Extended Descriptions

Masked Autoencoders (MAE). MAE [8] exemplifies the reconstruction paradigm. It randomly masks 75% of image patches, encodes only the visible ones, and trains a decoder to reconstruct missing pixels via $\mathcal{L}_{\text{MAE}} = \|x_{\text{masked}} - \hat{x}_{\text{masked}}\|^2$. This forces the encoder to retain enough information to reconstruct even noisy portions, making reconstruction methods sensitive to augmentation design.

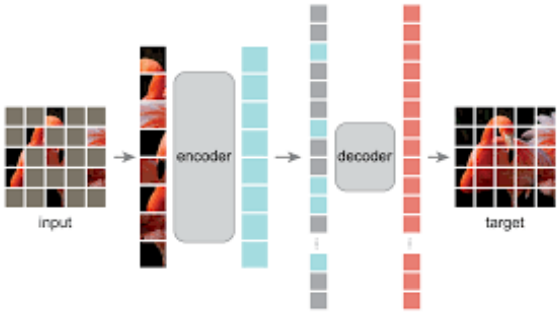

Figure 1: Main components of MAE architecture

SimCLR. SimCLR [3] constructs $2N$ augmented views from a batch of N images and trains via the NTXent loss:

$$\mathcal{L}_{\text{SimCLR}} = -\log \frac{\exp\big(\text{sim}(z_i, z_j)/\tau\big)}{\sum_{k \neq i} \exp\big(\text{sim}(z_i, z_k)/\tau\big)} \tag{2}$$

where $z_i = g_\phi(f_\theta(\tau(x)))$, $\text{sim}(\cdot, \cdot)$ is cosine similarity, and τ is a temperature. Positive pairs are pulled together while negatives are pushed apart.

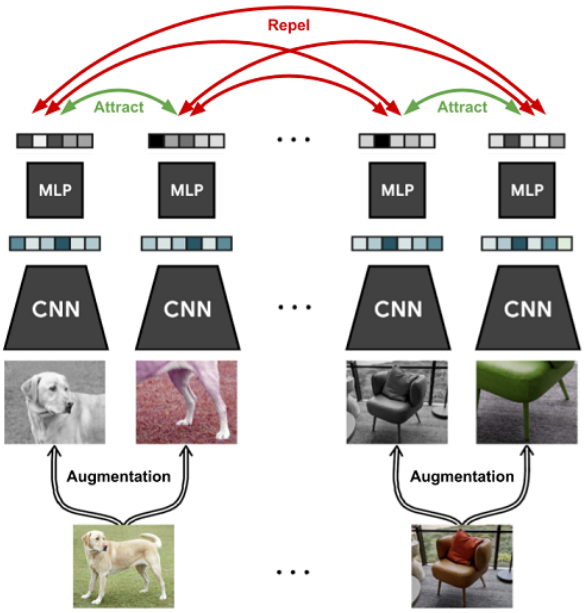

Figure 2: Main components of SimCLR architecture

VICReg. VICReg [4] avoids negative pairs by regularizing the embedding distribution:

$$\mathcal{L}_{\text{VICReg}} = \lambda \cdot \mathcal{L}_{\text{inv}} + \mu \cdot \mathcal{L}_{\text{var}} + \nu \cdot \mathcal{L}_{\text{cov}}. \tag{3}$$

The invariance term is MSE between view embeddings; the variance term prevents collapse via a hinge loss on per-dimension standard deviation; the covariance term decorrelates dimensions by penalizing off-diagonal entries of the covariance matrix.

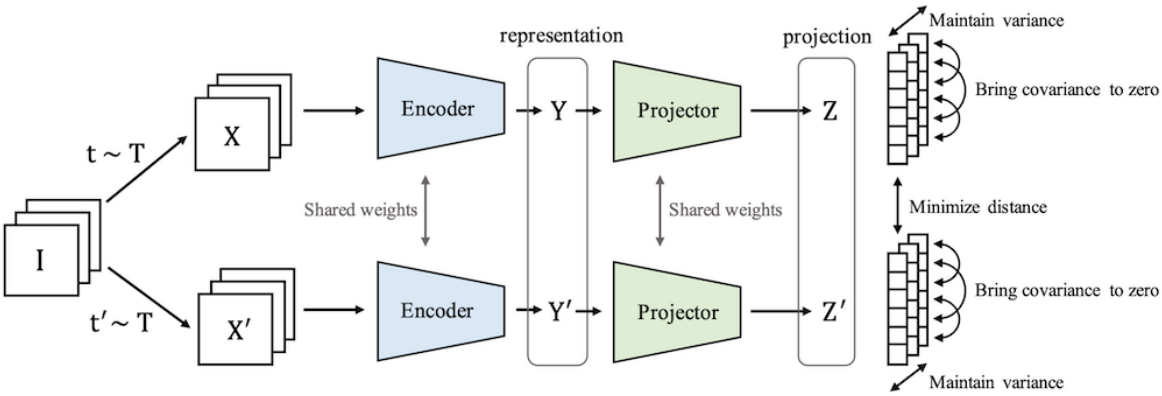

Figure 3: Main components of VICReg architecture

DINO. DINO [6] frames SSL as self-distillation: a teacher, updated via an exponential moving average of the student weights, processes global crops while the student sees local crops, learning to match the teacher's output distribution via cross-entropy.

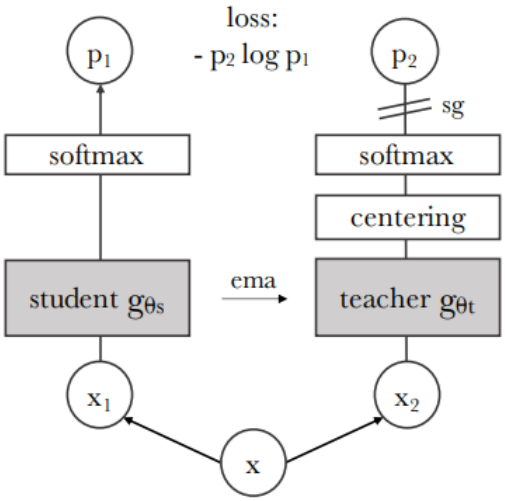

Figure 4: Main components of DINO architecture

LeJEPA. LeJEPA [7] derives its objective from a theoretical analysis of JEPA embeddings. It enforces isotropic Gaussian embeddings via Sketched Isotropic Gaussian Regularization (SIGReg):

$$\mathcal{L}_{\text{LeJEPA}} = \mathcal{L}_{\text{pred}} + \beta \cdot \mathcal{L}_{\text{SIGReg}} \tag{4}$$

where $\mathcal{L}_{\text{pred}}$ is a latent prediction loss. Operating entirely in latent space, it shares the robustness advantages of joint-embedding methods with a single hyperparameter β.

B Training and evaluation

Table 4: Hyperparameters and Architectural Details

Component	**Parameter**	**Configuration**
ViT Backbone	Input Resolution	64×64 (grayscale)
	Patch Size	8×8
	Layers / Heads	6 / 4
	Embedding Dimension	256
SSL Configurations	SimCLR & VICReg	Projector to latent space
	MAE	Decoder depth: 2, dim: 128, Mask ratio: 0.75
	LeJEPA	λ (SIGReg/invariance trade-off): 0.5
Pre-training	Optimizer	Adam
	Learning Rate / Batch Size	10^{-3} / 256
	Epochs	20
Evaluation	Protocol	Frozen linear probe
	Epochs	10
	Metric	Top-1 accuracy (Clean & Corrupted)